

ML Foundations — Problem Framing, Hold-Out Test & Baseline

Dataset: Adult / Census Income (UCI, via OpenML) | Qasim | random_state = 42

Problem Framing

We predict whether a household's income exceeds \$50K/year from 1994 U.S. Census attributes (age, education, occupation, hours worked, etc.). The business objective is marketing cost control: a targeted-offer campaign wants to identify likely high-income households instead of contacting everyone, reducing wasted outreach spend. The target is binary: income >50K = 1 (positive), ≤50K = 0 (negative). The positive class base rate in the full dataset is **23.93%**, so the dataset is moderately imbalanced and a naive "always predict low income" guess is already correct 76.07% of the time on accuracy alone.

Chosen Metric

Primary metric: ROC AUC. No fixed campaign budget or decision threshold has been set yet, so a threshold-independent measure of ranking quality is more useful than a metric tied to one cutoff (e.g. Precision@top-k). ROC AUC lets the marketing team later choose their own targeting threshold based on budget while still comparing models fairly today. PR AUC is tracked as a secondary check, since ROC AUC can look optimistic under class imbalance.

Baseline Results (held-out test set, n = 9,769, 20% stratified split)

Baseline	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
A: Majority class	0.7607	0.0000	0.0000	0.0000	0.5000	0.2393
B: Rule (education-num ≥ 13)	0.7530	0.4844	0.4970	0.4906	0.7189	0.4443

Baseline B (education-num ≥ 13, i.e. roughly a Bachelor's degree or higher) clearly outperforms Baseline A on every metric except raw accuracy, where majority-class prediction is inflated by class imbalance. Baseline A cannot identify any true positive by construction, so its precision/recall/F1/PR AUC are 0 and its ROC AUC (0.50) reflects zero ranking ability. Baseline B's ROC AUC of 0.72 shows that even a single sensible feature carries real signal. For a future model to be considered "useful" over these baselines, it should exceed Baseline B's ROC AUC by a clear margin — we set the bar at roughly a **5–10 percentage-point absolute improvement** in ROC AUC (comparable gains in F1/PR AUC), since a smaller gain could plausibly be split-to-split noise rather than a genuinely better model.

Error Analysis

Manually inspecting 10 sampled false positives and 10 false negatives from Baseline B (1,237 false positives and 1,176 false negatives total on the test set) surfaced six recurring patterns: (1) degree holders in lower-paying occupations (e.g. clerical, other-service) are false positives; (2) high earners without a formal degree — often working long hours or in skilled/managerial roles — are false negatives; (3) nonzero capital-gain, a strong income signal, is ignored by the rule; (4) married individuals are disproportionately false negatives, since marital status correlates with income but isn't used; (5) high-cardinality fields like occupation and native-country carry information a single rule can't exploit; and (6) hours-per-week appears to interact with education rather than acting independently.

Conclusion: Given Baseline A's ROC AUC of 0.50 and Baseline B's ROC AUC of 0.72, the priority for the rest of the week is to optimize **ROC AUC** using a multi-feature model (e.g. logistic regression or gradient-boosted trees), since the error analysis shows income depends on interactions between education, hours worked, occupation, marital status, and capital gains that no single-feature rule can capture. The target is a ≥5–10 point ROC AUC improvement over Baseline B (0.72) before the added model complexity is considered worth deploying.